Do Now · retrieval

Thinking ahead: inputs, caching and reuse

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why omit cover colour from a simple book-loan model?

Q2

What is a reusable subroutine?

Q3

Calculate 7×7.

Q4

Why is caching squares safer than caching changing attendance data?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Define the contract — The input is an integer, and the output is its square; overflow is a possible precondition concern.

Core idea 2

Trace repeated work — First call for 5 calculates and caches 25; second returns the stored result.

Core idea 3

Evaluate caching — Saves repeated calculation but costs memory. For changing data, cached values can become out of date.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

input — Data passed into a computation.

output — Result of a computation.

precondition — Condition required before a process can run correctly.

cache hit — Requested value found in storage.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

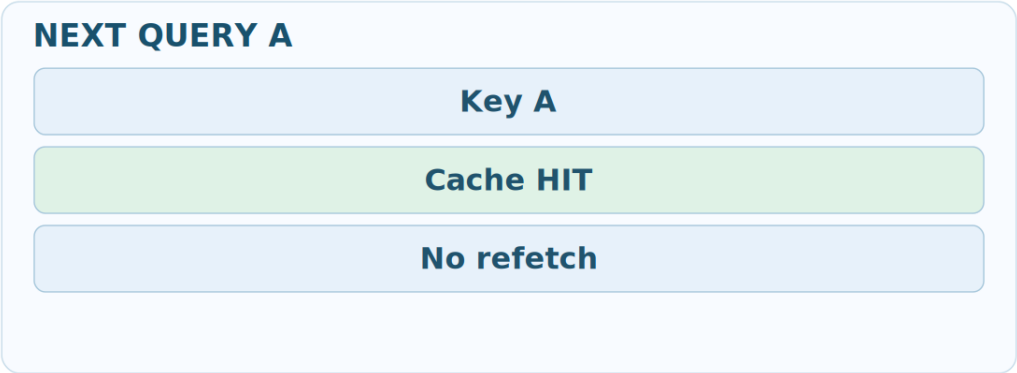
Problem

A cached ticket price for zone A is £3.40. The same zone is requested twice.

Follow the worked method

1. Predict first — The key A is already stored, so the lookup uses the cache.
2. Model and check — If pricing changes, the cached price must be updated or expired.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Using the old cached 12 gives a stale answer instead of the new total 15.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For SquareCached(6), identify input, output and two preconditions or assumptions.

Question 2

Trace two calls of SquareCached(7) starting with an empty cache.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Give three advantages/disadvantages of caching changing school data and a way to avoid stale values.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Once a cached value is stored it is always correct.

Use key vocabulary from memory

1. Explain input in your own words.

2. Explain output in your own words.

3. Explain precondition in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.